

# FITBUDDY

AI-Powered Personalized Workout and Wellness Assistant

## PHASE 03

### Project Design Phase

*Define the architecture, components, data model, AI contracts, API routes, and interface design.*

Prepared for phase-wise submission  
30 September 2026

**Phase objective:** Define the architecture, components, data model, AI contracts, API routes, and interface design.

## 1. High-Level Architecture

- Browser / Jinja2 templates -> FastAPI routes -> service layer -> SQLite database and Gemini client.
- Static CSS and JavaScript provide presentation and lightweight client-side interaction.
- Pydantic schemas act as the contract for both user input and structured AI output.

## 2. Component Design

- `app/main.py`: application entry point, static file mounting, startup database initialization.
- `app/routes.py`: HTTP routes, form handling, database operations, and HTML responses.
- `app/schemas.py`: validation and structured output models.
- `app/models.py`: SQLAlchemy User and Plan entities.
- `app/database.py`: engine, session factory, database initialization, dependency injection.
- `app/services/*`: AI client, workout generation, nutrition generation, and feedback revision.
- `templates/*` and `static/*`: user interface.
- `tests/*`: automated verification.

## 3. Data Model

- User entity: `id`, `user_id`, `username`, `age`, `weight`, `goal`, `intensity`, `created_at`.
- Plan entity: `id`, `user_id`, `original_plan`, `updated_plan`, `nutrition_tip`, `feedback`, `created_at`, `updated_at`.
- Relationship: one User has at most one Plan, with cascade deletion.

## 4. AI Design

- Workout output uses the `WorkoutPlan` schema: `title`, `summary`, `safety_note`, and exactly seven `WorkoutDay` objects.
- Each `WorkoutDay` contains `day`, `focus`, `warmup`, `exercises`, `cooldown`, and optional `recovery_note`.
- Nutrition output uses `NutritionTip`: `tip`, `hydration`, `recovery`.
- AI responses are requested as JSON and validated using the same Pydantic schemas before persistence.
- Mock generators provide deterministic local behavior for demonstrations and tests.

## 5. API and Navigation Design

- `GET /`: home page.
- `POST /generate-workout`: create/update user and generate plan.
- `POST /submit-feedback`: revise a saved plan.
- `GET /view-all-users`: local user dashboard.
- `POST /delete-user/{user_id}`: delete a local user.
- `GET /health`: health response.
- `GET /docs`: Swagger UI.

## 6. Interface Design

- Home page: collect profile information.
- Result page: present plan and nutrition output; provide feedback flow.
- All-users page: show stored local records.
- Error page: present validation or AI service errors in a simple format.
- Responsive CSS is used for browser-friendly presentation.

## Design Contracts

| Contract     | Key rule                                                 |
|--------------|----------------------------------------------------------|
| UserInput    | Age 13-100; weight >20; controlled goal/intensity values |
| WorkoutPlan  | Exactly 7 days                                           |
| WorkoutDay   | At least 1 exercise; warm-up and cooldown required       |
| Exercise     | Sets/ reps or duration where appropriate                 |
| NutritionTip | Tip + hydration + recovery                               |

## Phase Deliverables / Evidence

- Architecture and component design
- Database and AI schema design
- API and UI design contracts

## Submission Note

This phase document is intended to be submitted together with the corresponding FitBuddy source project and evidence captured during execution. Keep API keys, personal credentials, and other secrets out of screenshots and submissions.